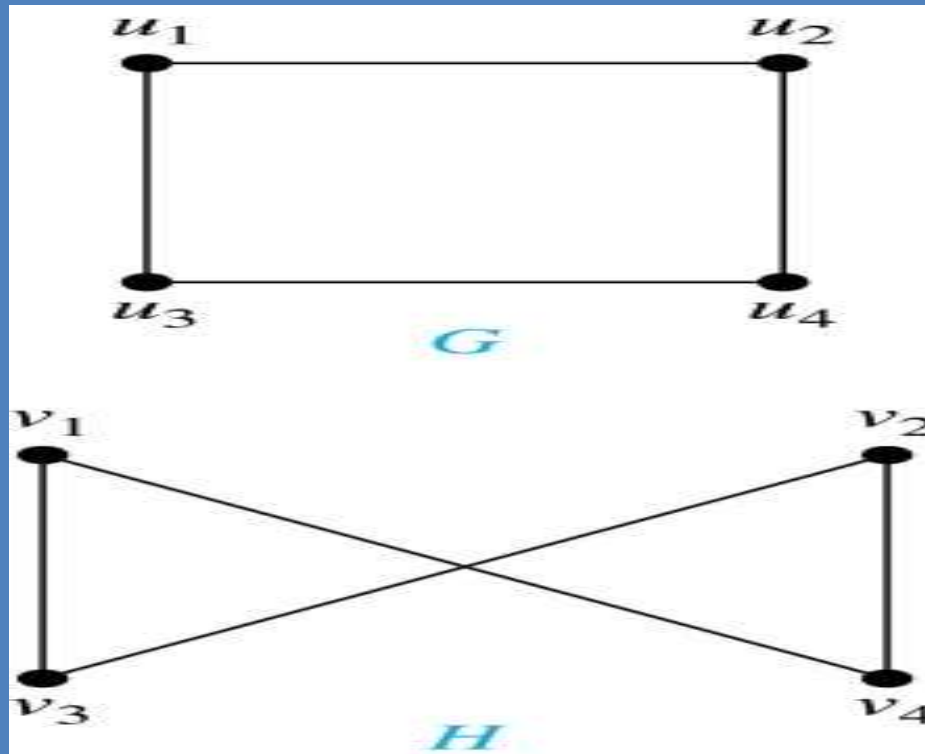


Isomorphism of Graphs

Definition: The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are *isomorphic* if there is a *bijjective (one-to-one and onto)* function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all a and b in V_1 . Such a function f is called an *isomorphism*. Two simple graphs that are not isomorphic are called *non isomorphic*.

Example: Show that the graphs $G = (V, E)$ and $H = (W, F)$ are isomorphic.



Solution: The function f with $f(u_1) = v_1$, $f(u_2) = v_4$, $f(u_3) = v_3$, and $f(u_4) = v_2$ is a one-to-one correspondence between V and W . Note that adjacent vertices in G are u_1 and u_2 , u_1 and u_3 , u_2 and u_4 , and u_3 and u_4 . Each of the pairs $f(u_1) = v_1$ and $f(u_2) = v_4$, $f(u_1) = v_1$ and $f(u_3) = v_3$, $f(u_2) = v_4$ and $f(u_4) = v_2$, and $f(u_3) = v_3$ and $f(u_4) = v_2$ consists of two adjacent vertices in H .

Isomorphism of Graphs (*cont.*)

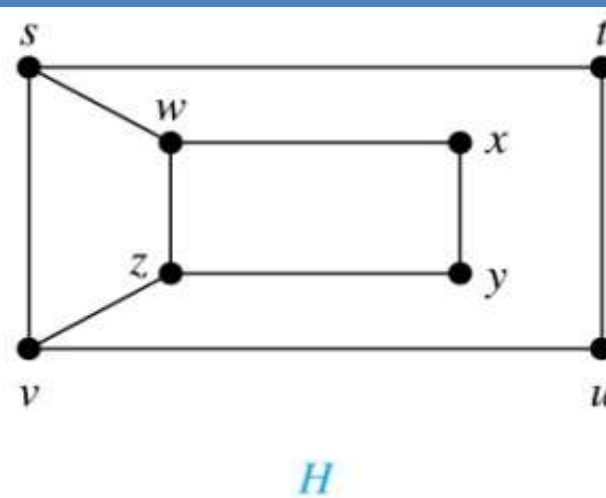
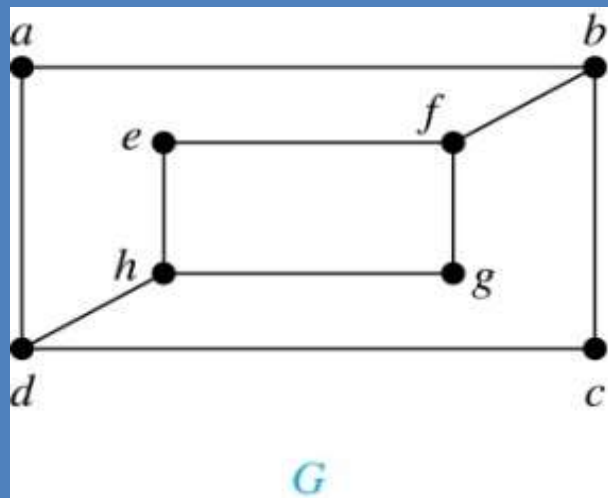
- It is difficult to determine whether two simple graphs are isomorphic using brute force because there are $n!$ possible one-to-one correspondences between the vertex sets of two simple graphs with n vertices.
- The best algorithms for determining whether two graphs are isomorphic have exponential worst case complexity in terms of the number of vertices of the graphs.

Isomorphism of Graphs (*cont.*)

- Sometimes it is not hard to show that two graphs are not isomorphic. We can do so by finding a property, preserved by isomorphism, that only one of the two graphs has. Such a property is called *graph invariant*.
- There are many different useful graph invariants that can be used to distinguish nonisomorphic graphs, such as the **number of vertices**, **number of edges**, and **degree sequence** (list of the degrees of the vertices in nonincreasing order).

Isomorphism of Graphs (*cont.*)

Example: Determine whether these two graphs are isomorphic.



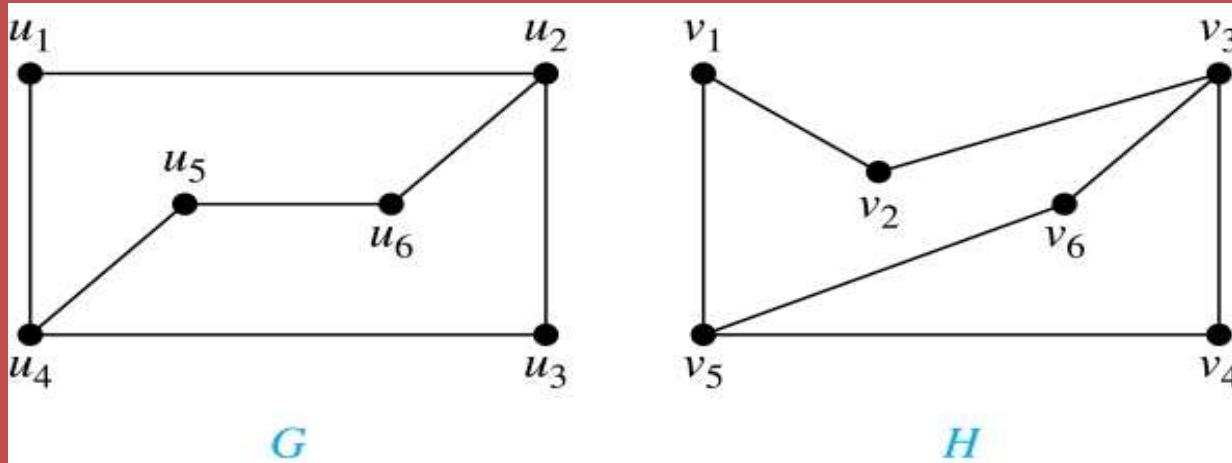
Isomorphism of Graphs (*cont.*)

Solution: Both graphs have eight vertices and ten edges.

They also both have four vertices of degree two and four of degree three.

However, G and H are not isomorphic. Note that since $\deg(a) = 2$ in G , a must correspond to t, u, x , or y in H , because these are the vertices of degree 2. But each of these vertices is adjacent to another vertex of degree two in H , which is not true for a in G .

Example: Determine whether these two graphs are isomorphic.



Solution: Both graphs have six vertices and seven edges. They also both have four vertices of degree two and two of degree three.

The subgraphs of G and H consisting of all the vertices of degree two and the edges connecting them are isomorphic. So, it is reasonable to try to find an isomorphism f .

We define an injection f from the vertices of G to the vertices of H that preserves the degree of vertices. We will determine whether it is an isomorphism.

The function f with $f(u_1) = v_6$, $f(u_2) = v_3$, $f(u_3) = v_4$, and $f(u_4) = v_5$, $f(u_5) = v_1$, and $f(u_6) = v_2$ is a one-to-one correspondence between G and H . Showing that this correspondence preserves edges is straightforward, so we will omit the details here. Because f is an isomorphism, it follows that G and H are isomorphic graphs.